Trastuzumab deruxtecan in patients in the USA and Europe with HER2-positive advanced gastric or gastroesophageal junction cancer with disease progression on or after a trastuzumab-containing regimen (DESTINY-Gastric02): primary and updated analyses from a single-arm, phase 2 study

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Summary

Background Approximately 15–20% of advanced gastric and gastro-oesophageal junction cancers overexpress HER2. In DESTINY-Gastric01, the HER2-targeted antibody-drug conjugate trastuzumab deruxtecan improved response and overall survival versus chemotherapy in patients from Japan and South Korea with locally advanced or metastatic HER2-positive gastric or gastro-oesophageal junction cancer whose disease progressed after two lines of previous therapy including trastuzumab. Here, we report primary and updated analyses of the single-arm, phase 2 DESTINY-Gastric02 trial, which aimed to examine trastuzumab deruxtecan in patients living in the USA and Europe.

Methods DESTINY-Gastric02 is a single-arm, phase 2 study in adult patients from 24 study sites in the USA and Europe (Belgium, Spain, Italy, and the UK). Eligible patients were aged at least 18 years and had an Eastern Cooperative Oncology Group performance status of 0 or 1, pathologically documented unresectable or metastatic gastric or gastro-oesophageal junction cancer, progressive disease on or after first-line therapy with a trastuzumab-containing regimen, with at least one measurable lesion per Response Evaluation Criteria in Solid Tumours (version 1.1), and centrally confirmed HER2-positive disease on a postprogression biopsy. Patients were given 6·4 mg/kg of trastuzumab deruxtecan intravenously every 3 weeks until disease progression, withdrawal by patient, physician decision, or death. The primary endpoint was confirmed objective response rate by independent central review. The primary endpoint and safety were assessed in the full analysis set (ie, participants who received at least one dose of study drug). Here, we report the primary analysis of this study, with a data cutoff of April 9, 2021, and an updated analysis, with a data cutoff of Nov 8, 2021. This trial is registered with ClinicalTrials.gov, NCT04014075, and is ongoing.

Findings Between Nov 26, 2019, and Dec 2, 2020, 89 patients were screened and 79 were enrolled and subsequently treated with trastuzumab deruxtecan (median age 60·7 years [IQR 52·0–68·3], 57 [72%] of 79 were male, 22 [28%] were female, 69 [87%] were White, four [5%] were Asian, one [1%] was Black or African American, one [1%] was Native Hawaiian or Pacific Islander, one had missing race, and three [4%] were other races). At the primary analysis (median follow-up 5·9 months [IQR 4·6–8·6 months]), confirmed objective response was reported in 30 (38% [95% CI 27·3–49·6]) of 79 patients, including three (4%) complete responses and 27 (34%) partial responses, as assessed by independent central review. As of data cutoff for the updated analysis (median follow-up 10·2 months [IQR 5·6–12·9]), a confirmed objective response was reported in 33 (42% [95% CI 30·8–53·4]) of 79 patients, including four (5%) complete responses and 29 (37%) partial responses, as assessed by independent central review. The most common grade 3 or worse treatment-emergent adverse events were anaemia (11 [14%]), nausea (six [8%]), decreased neutrophil count (six [8%]), and decreased white blood cell count (five [6%]). Drug-related serious treatment-emergent adverse events occurred in ten patients (13%). Deaths determined to be associated with study treatment occurred in two patients (3%) and were due to interstitial lung disease or pneumonitis.

Interpretation These clinically meaningful results support the use of trastuzumab deruxtecan as second-line therapy in patients with HER2-positive advanced gastric or gastro-oesophageal junction cancer.

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Research in context

Evidence before this study

We searched PubMed for clinical trials evaluating second-line treatments for HER2-positive gastric or gastro-oesophageal junction cancer published between July 1, 2015, and April 30, 2022. The search terms included were "HER2 positive" OR "HER2 expressing" OR "HER2 amplified" AND "gastric cancer" OR "gastroesophageal junction cancer"; the search was restricted to English language publications. The results of this search were used to put our findings from the DESTINY-Gastric02 study into context in terms of its objectives and findings. The HER2-targeted antibody-drug conjugate trastuzumab deruxtecan (DS-8201) is approved in several countries worldwide for the treatment of patients with locally advanced or metastatic gastric or gastro-oesophageal junction adenocarcinoma who have previously received a trastuzumab-based regimen. Approvals were based on the phase 2, open-label, multicentre, randomised DESTINY-Gastric01 trial of trastuzumab deruxtecan in patients from Japan and South Korea.

Added value of this study

This is the first phase 2 study of trastuzumab deruxtecan in patients from the USA and Europe with unresectable or metastatic gastric or gastro-oesophageal junction cancer who have had disease progression on or after a trastuzumab-containing regimen. Trastuzumab deruxtecan exhibited clinical activity in this patient population and a safety profile generally consistent with that observed in previous studies.

Implications of all the available evidence

These results support further clinical research into trastuzumab deruxtecan, including the ongoing phase 3 DESTINY-Gastric04 trial, which will investigate the efficacy and safety of trastuzumab deruxtecan compared with ramucirumab and paclitaxel in patients with gastric or gastro-oesophageal junction cancer whose disease has progressed on or after a trastuzumab-containing regimen.

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Introduction

Approximately 15–20% of advanced gastric and gastro-oesophageal junction cancers overexpress HER2 (also known as ERBB2).[1] Recommended first-line treatment for patients with HER2-positive gastric or gastro-oesophageal junction cancer is chemotherapy in combination with the anti-HER2 antibody trastuzumab.[2] Results from the phase 3 KEYNOTE-811 study[3] supported accelerated approval by the US Food and Drug Administration of pembrolizumab in combination with trastuzumab and chemotherapy for first-line treatment of locally advanced unresectable or metastatic HER2-positive gastric or gastro-oesophageal junction cancer.[4] Key second-line trials of HER2-targeted therapy in patients with HER2-positive gastric or gastro-oesophageal junction cancer did not meet their primary endpoint of overall survival,[5,6] with evidence suggesting that a loss of HER2 expression after first-line trastuzumab therapy might be a mechanism contributing to resistance to subsequent lines of anti-HER2 treatment.[7]

Trastuzumab deruxtecan has been approved in multiple countries worldwide for use in the treatment of adult patients with locally advanced or metastatic HER2-positive gastric or gastro-oesophageal junction adenocarcinoma who have previously received a trastuzumab-based regimen.[8–10] Trastuzumab deruxtecan is an antibody–drug conjugate composed of a humanised anti-HER2 monoclonal antibody linked to a topoisomerase I inhibitor payload through a tetrapeptide-based cleavable linker.[11–13] After being internalised by tumour cells, the linker is cleaved by lysosomal enzymes that are upregulated in those cells. The cytotoxic payload, which is cell membrane-permeable after cleavage, has a bystander antitumour effect that enables the killing of tumour cells proximal to HER2-expressing tumour cells.[11–13]

The efficacy and safety of trastuzumab deruxtecan in gastric or gastro-oesophageal junction adenocarcinoma was established in the open-label, multicentre, randomised, phase 2 DESTINY-Gastric01 trial[14] of trastuzumab deruxtecan in patients from Japan and South Korea with HER2-positive, locally advanced or metastatic gastric or gastro-oesophageal junction adenocarcinoma that had progressed after at least two previous lines of therapy, including a fluoropyrimidine agent, a platinum agent, and trastuzumab.[8–10] In DESTINY-Gastric01, trastuzumab deruxtecan showed significant benefit in objective response rate and overall survival versus physician's choice of standard chemotherapy.[14] A range of factors, including diet, smoking status, and infection with *Helicobacter pylori*, might contribute to differences in outcomes for Asian and North American and European populations of patients with gastric cancer.[15,16] Gastric cancer is diagnosed earlier in many Asian countries because of robust screening guidelines in comparison with many other regions; this disparity results in generally worse prognoses and 5-year survival rates for patients with gastric cancer in countries in Europe and the USA.[15]

Here we report the primary and updated results from the DESTINY-Gastric02 study. The objective of this study was to assess the activity and safety of trastuzumab deruxtecan monotherapy in patients in Europe and the USA with metastatic or unresectable HER2-positive gastric or gastro-oesophageal junction cancer whose disease had progressed on or after a trastuzumab-containing regimen and were confirmed to have retained tumour HER2 positivity.

Methods

Study design and participants

See Online for appendix

DESTINY-Gastric02 is a multicentre, single-arm, phase 2 study that recruited patients from 24 hospitals, academic centres, and specialist cancer centres in the USA, Belgium, Spain, Italy, and the UK (appendix pp 3–4). Eligible patients were aged at least 18 years and had pathologically documented, unresectable or metastatic gastric or gastro-oesophageal junction cancer, progressive disease on or after first-line therapy with a trastuzumab-containing regimen, centrally confirmed HER2-positive disease (defined as immunohistochemistry 3+ or immunohistochemistry 2+ and evidence of HER2 amplification by in situ hybridisation on a fresh post-trastuzumab progression biopsy), an Eastern Cooperative Oncology Group (ECOG) performance status score of 0 or 1, a left ventricular ejection fraction of at least 50%, and adequate organ function. Patients must have had at least one measurable lesion as per Response Evaluation Criteria in Solid Tumours (RECIST) version 1.1. All lesions (target and non-target) were measured at screening according to RECIST version 1.1.

The post-trastuzumab progression biopsy sample could be obtained from a primary or metastatic tumour site by means of standard biopsy per local practice or a surgical sample. To account for possible heterogeneity of HER2 expression in tumour tissue, the study protocol specified that multiple core biopsies (at least two) be taken. Needle biopsies of fluid or cells and biopsies derived from bone (metastatic) were not acceptable. Previous adjuvant therapy with a trastuzumab-containing regimen was counted as a line of therapy if progression occurred within 6 months of completing adjuvant therapy. Exclusion criteria included spinal cord compression or clinically active central nervous system metastases, uncontrolled or significant cardiovascular disease, clinically severe pulmonary compromise, pleural effusion, ascites, or pericardial effusion, or a history of non-infectious interstitial lung disease (ILD) or pneumonitis that required corticosteroid therapy, current ILD or pneumonitis, or were suspected to have ILD or pneumonitis that could not be ruled out by imaging at screening. Additionally, patients were excluded if they used anticancer therapy after first-line treatment with a trastuzumab-containing regimen. A full list of eligibility criteria is in the protocol (appendix).

The study was done in accordance with the principles outlined in the Declaration of Helsinki and the International Conference of Harmonisation Guidelines for Good Clinical Practice and the protocol was approved by the independent ethics committee or institutional review board at each site. Written informed consent was obtained from all patients before initiation of the study.

Procedures

All patients were given 6·4 mg/kg of trastuzumab deruxtecan intravenously every 3 weeks until disease progression as per RECIST version 1.1; clinical progression; or withdrawal by patient, physician decision, or death. Each patient had a 40-day (up to 47 days) follow-up visit after the last study drug administration or before starting new anticancer treatment. Long-term survival follow-up visits occurred every 12 weeks (±14 days) from the date of the 40-day (up to 47 days) follow-up visit, until death, withdrawal of consent, or loss to follow-up, whichever occurred first. Sex or gender and race or ethnicity were captured through the electronic data capture system.

Tumour assessments of the chest, abdomen, pelvis, and any other sites of disease via CT or MRI scans were done within 28 days of the first dose, every 6 weeks (±7 days) starting on cycle 1 day 1 for the first year of treatment, every 12 weeks (±7 days) after the first year until disease progression or start of a new anticancer treatment, and at the 40-day (up to 47 days) follow-up visit if tumour assessment was not done at the end of treatment visit or if study treatment was discontinued for any reason other than disease progression. A CT or MRI of the brain was done for all patients within 28 days before recruitment and every 6 weeks (±7 days) from cycle 1 day 1 if the patients had stable brain metastases at baseline. The brain CT or MRI scan was done as clinically indicated for all other patients. Activity endpoints were based on tumour assessments done at screening and every 6 weeks during the first year of treatment and every 12 weeks thereafter until objective disease progression, based on radiological assessment, or the initiation of a new anticancer therapy. Tumour scans were assessed by independent central review and by local investigators.

Laboratory monitoring was done during screening, within 14 days before first dose of study drug, within 3 days before the infusion for all cycles, at cycle 1 day 8 and cycle 1 day 15 (±1 days), at end of treatment (defined as the date the investigator decided to discontinue study treatment [+7 days]), and at the 40-day (up to 47 days) follow-up visit. Blood samples for biomarker analyses were taken within 3 days before the administration of trastuzumab deruxtecan and on cycle 1 day 1, on day 1 of every three cycles (ie, cycles 4, 7, and so on), and at the end of treatment.

At each study visit, patients were evaluated for adverse events, which were directly observed, reported spontaneously by the patient, or established through questioning of the patient. Treatment-emergent adverse events were assessed and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0. Treatment-emergent adverse events were defined as an adverse event that was absent before the first dose of study drug or that had worsened in severity or seriousness after initiating study drug until 47 days after the last dose. Treatment-emergent adverse events also included serious adverse events with an onset or worsening 48 days or more after last dose of the study drug, if related to the study drug.

Suspected ILD or pneumonitis cases were adjudicated by an external independent adjudication committee. For events of ILD or pneumonitis, established ILD and pneumonitis management guidelines recommend to carefully monitor for signs or symptoms (cough, fever, and dyspnoea); discontinue, interrupt, or reduce the dose of trastuzumab deruxtecan; conduct imaging; and start corticosteroid therapy as soon as ILD–pneumonitis was suspected.[17]

Two dose reductions of trastuzumab deruxtecan were allowed, from 6·4 mg/kg to 5·4 mg/kg and then 4·4 mg/kg; dose increases were not permitted after reductions. Patients with more than two dose reductions were withdrawn from the study treatment if further adverse events meeting the requirement for dose reductions occurred. A dose could be delayed for up to 28 days (or 49 days from the last infusion date) from the planned date of the administration, based on investigator discretion and by dose modification guidelines. Patients who required a dose delay longer than 28 days were withdrawn from the study. Administration of subsequent anticancer treatment was based on investigator discretion, and data on subsequent anticancer treatments were collected.

Outcomes

The primary endpoint was confirmed objective response rate, defined as the proportion of patients who had a best overall response of confirmed complete response or partial response, per RECIST version 1.1, assessed by independent central review and confirmed with a second assessment at least 4 weeks from the time a response was first acquired. The key secondary endpoint was progression-free survival, defined as the time from the date of first dose of study drug to the date of radiographic disease progression per RECIST version 1.1, or death due to any cause assessed by independent central review. Other secondary endpoints were overall survival (defined as the time interval from the date of first dose of study drug to the date of death due to any cause), progression-free survival by investigator assessment per RECIST version 1.1, objective response rate by investigator assessment per RECIST version 1.1, duration of response (defined as the time from the date of first documentation of objective response to the date of the first documentation of disease progression or death due to any cause) assessed by independent central review and investigator, and safety. Safety endpoints include serious adverse events, treatment-emergent adverse events, drug-related treatment-emergent adverse events, discontinuations owing to adverse events, and adverse events of special interest. Prespecified exploratory endpoints included disease control rate (defined as complete response rate plus partial response rate plus stable disease rate) by independent central review and investigator assessment, time to response by independent central review and investigator assessment, best percentage change in tumour size from baseline by independent central review (sum of diameters of measurable tumours calculated based on RECIST 1.1 criteria, calculated as the percent change in the smallest sum of diameters from all post-baseline tumour assessments compared to baseline), and assessment of exploratory biomarkers, which will be reported elsewhere. As only selected efficacy outputs were requested by health authorities for the updated data cutoff, and central review assessments are considered to provide a more accurate imaging review, PFS by investigator was therefore not analysed at the updated data cutoff.

Statistical analysis

On the basis of a reported best objective response rate of approximately 27% in second-line treatment of patients from Europe, Israel, Australia, and the USA with gastric or gastro-oesophageal junction cancer,[18] a sample size of 72 patients would be needed to provide 90% power to achieve a lower limit of 95% CI for the objective response rate exceeding 27% under the expected objective response rate of 45%. The study was planned to enrol an additional 10% of patients to account for potential withdrawals, for an estimated sample size of 80 patients.

Primary and secondary activity analyses as well as safety analyses were done in the full analysis set, which included all participants who were enrolled and received at least one dose of the study drug. Exploratory analyses were done on the basis of the full analysis set and the availability of assessment. For response assessments, patients with missing baseline or post-baseline assessments were included in the analysis set but were not counted as responders.

The primary analysis of confirmed objective response rate by independent central review was done for the full

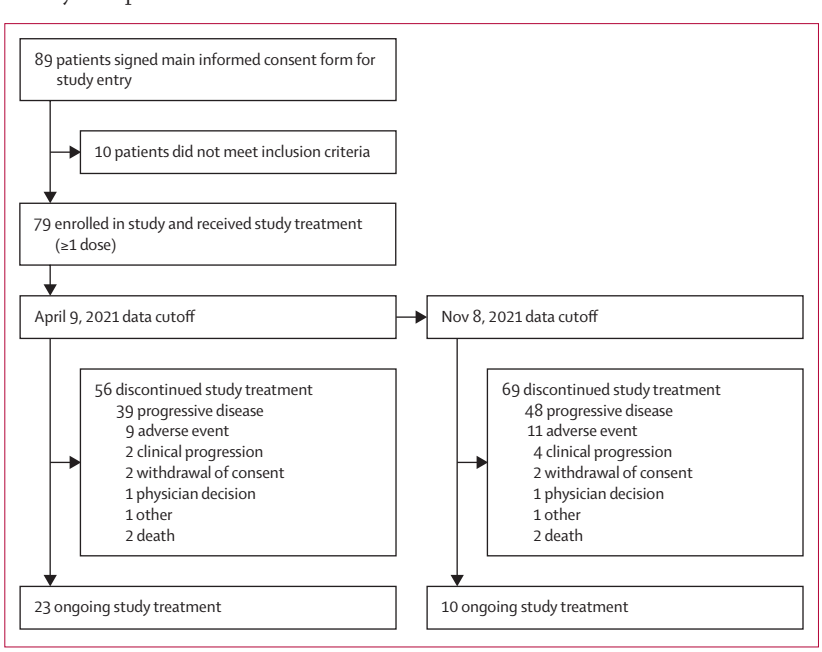


Figure 1: **Trial profile**

	Patients (N=79)
Age	
Median, years	60·7 (52·0–68·3)
<65	46 (58%)
≥65	33 (42%)
Sex	
Male	57 (72%)
Female	22 (28%)
Race	
White	69 (87%)
Black or African American	1 (1%)
Asian	4 (5%)
Native Hawaiian or Pacific Islander	1 (1%)
Other	3 (4%)
Missing	1 (1%)
Eastern Cooperative Oncology Group performance status	
0	29 (37%)
1	50 (63%)
HER2 expression	
IHC 3+	68 (86%)
IHC 2+ and ISH+	10 (13%)
Not evaluable	1 (1%)*
Adenocarcinoma subtype	78 (99%)†
Intestinal	19 (24%)
Diffuse	1 (1%)
Mixed	1 (1%)
Unknown	57 (72%)‡
Cancer type	
Gastric	27 (34%)
Gastro-oesophageal junction	52 (66%)
Disease presentation at initial diagnosis	
Metastatic disease or locally advanced	69 (87%)
Locoregional disease	9 (11%)
Unknown	1 (1%)

(Table 1 continues in next column)

	Patients (N=79)
(Continued from previous column)	
Number of metastatic sites	
<2	5 (6%)
≥2	74 (94%)
Previous total gastrectomy	
Yes	0
No	79 (100%)
Liver metastasis at baseline	50 (63%)
Time from diagnosis, months	14·2 (10·2–25·9)
Lines of previous systemic therapy intended for locally advanced or metastatic cancer	
0	6 (8%)§
1	73 (92%)
Previous platinum-based therapy	79 (100%)
Previous trastuzumab therapy	79 (100%)
Previous immune checkpoint inhibitor treatment	7 (9%)¶
Avelumab	1 (1%)
Pembrolizumab	6 (8%)

Data are median (IQR) or n (%). Demographics and baseline characteristics were not expected to change at the updated data cutoff and were not re-analysed; data are reported based on the April 9, 2021, data cutoff. IHC=immunohistochemistry. ISH=in-situ hybridisation. *Patient enrolment was based on local laboratory testing. †One patient had an intramucosal adenocarcinoma subtype and was not included in the 78 patients recorded as having adenocarcinoma. ‡Patients were classified as having unknown adenocarcinoma subtype by the investigator as part of the case report form data entry. §These patients were included in this study because they had experienced progression after receiving adjuvant or neoadjuvant therapy. ¶The number of patients who had received previous immune checkpoint inhibitor therapy might have been greater, because seven patients (9%) had previously participated in trials in which they were randomly assigned to double-blind treatment with either a checkpoint inhibitor or placebo.

Table 1: **Baseline characteristics**

analysis set when the last participant had at least 18 weeks of follow-up. Objective response rate estimate and its two-sided 95% exact CI were calculated with the Clopper-Pearson method. We did prespecified subgroup analyses of objective response rate according to region (North America *vs* Europe), age (<65 *vs* ≥65 years and <75 *vs* >75 years), sex (female *vs* male), ECOG PS (0 *vs* 1), HER2 status (IHC3+ *vs* IHC2+ and ISH+), primary tumour location (gastric *vs* gastro-oesophageal junction), histologic subtype (intestinal *vs* diffuse *vs* other), number of metastatic sites (two *vs* two or more), previous adjuvant or neoadjuvant therapy (yes *vs* no), previous nivolumab or pembrolizumab treatment (yes *vs* no), previous treatment with immune-checkpoint inhibitors or other immuno-oncology treatment (yes *vs* no), presence of liver metastases at baseline (yes *vs* no), renal impairment at baseline (normal *vs* mild *vs* moderate), and hepatic impairment at baseline (normal *vs* mild). For the progression-free survival analysis, patients who were alive with no objective documentation of radiological disease progression as assessed by independent central review at the time of data cutoff were censored at the date of their last evaluable tumour assessment. For the overall survival analysis, patients who were alive at the time of data analysis were censored at the last contact date at which the patient was known to be alive. For the duration of response analysis, patients who were alive and progression-free at the time of the analyses were censored at the date of the last evaluable tumour assessment. Kaplan-Meier estimates and survival curves are presented. For progression-free survival and duration of response, 95% CIs were calculated with the Brookmeyer-Crowley method. Duration of response was measured for responding patients (ie, those with a partial response or complete response) and was analysed on the basis of tumour assessment by independent central review. For time to response, median and 95% CIs were calculated on the basis of Kaplan-Meier estimates. Safety endpoints were assessed using descriptive statistics.

All analyses were done with using SAS version 9.4. The study is registered with ClinicalTrials.gov, NCT04014075.

Role of the funding source
The funder of the study Daiichi Sankyo had a role in study design, study oversight, data collection, data analysis, data interpretation, writing of the report, reviewing the manuscript, and the decision to submit for publication. The funder of the study AstraZeneca had a role in study oversight, data collection, data interpretation, writing of the report, reviewing the manuscript, and the decision to submit for publication.

Results

Between Nov 26, 2019, and Dec 2, 2020, 89 patients were screened, of whom 79 were enrolled and treated with at least one dose of trastuzumab deruxtecan (figure 1). Ten patients did not meet screening criteria after central HER2 confirmation and so were excluded. Patients had a median age of 60·7 years (IQR 52·0–68·3), 57 patients (72%) were male, 22 (28%) were female, 69 (87%) were White, four (5%) were Asian, one (1%) was Black or African American, one (1%) was Native Hawaiian or Pacific Islander, one (1%) had missing race data, and three (4%) were other races. Baseline demographics and disease characteristics are presented in table 1.

As of the April 9, 2021, primary analysis data cutoff (median follow-up of 5·9 months [IQR 4·6–8·6 months]), 30 patients (38%; 95% CI 27·3–49·6) had a confirmed objective response, as assessed by independent central review, with three (4%) having a complete response and 27 (34%) having a partial response. Additional antitumour activity outcomes are presented in table 2.

An updated analysis was requested as part of a health authority review (data cutoff, Nov 8, 2021). As of the updated analysis, ten (13%) patients remained on treatment. 39 patients received subsequent anticancer treatment after completion of study drug (appendix p 5).

As of the Nov 8, 2021, data cutoff (median follow-up of 10·2 months [IQR 5·6–12·9]), 33 patients (42%; 95% CI 30·8–53·4) had a confirmed objective response, as assessed by independent central review, with four (5%) having a complete response and 29 (37%) having a partial response. A summary of antitumour activity outcomes is presented in table 2. Best percentage change of tumour size from baseline by independent central review is shown in figure 2A. Subgroup analyses of objective response are shown in figure 2B. Median duration of response and median progression-free survival by independent central review, and overall survival, are shown in table 2 and figure 3. Investigator assessments of activity are shown in the appendix (p 6).

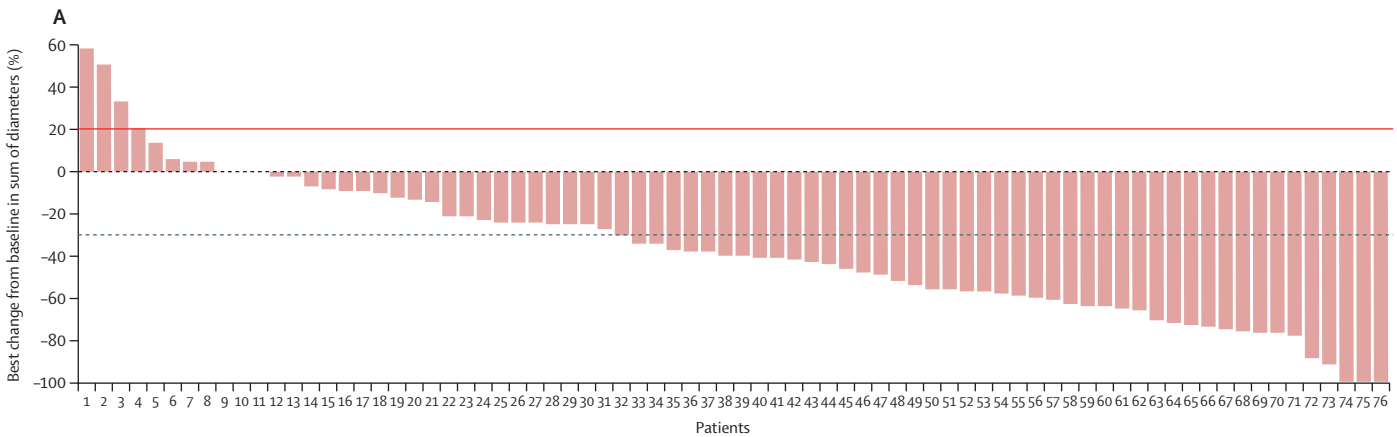
Median treatment duration was 4·3 months (IQR 2·7–10·1) as of the updated data cutoff (Nov 8, 2021). 79 (100%) patients had a treatment-emergent adverse event, 44 (56%) had a grade 3 or worse treatment-emergent adverse event, and 33 (42%) had a serious treatment-emergent adverse event (appendix p 7). Drug-related treatment-emergent adverse events were reported in 75 patients (95%), with 24 (30%) having grade 3 or worse adverse events and ten (13%) having serious adverse events. The most common grade 3 or worse treatment-emergent adverse events were anaemia (11 [14%]), nausea (six [8%]), decreased neutrophil count (six [8%]), and decreased white blood cell count (five [6%]; table 3). Treatment-emergent adverse events that led to study drug discontinuations occurred in 15 patients (19%; ten [13%] were drug related), that led to dose reductions occurred in 17 (22%; 14 [18%] were drug related), and that led to dose interruptions occurred in 23 (29%; eight [10%] were drug related; appendix p 7). Adjudicated ILD or pneumonitis was the most common drug-related treatment-emergent adverse event associated with treatment discontinuation (eight patients [10%]). Nausea (six [8%]) and decreased neutrophil count (four [5%]) were the most common drug-related treatment-emergent adverse events associated with dose reduction. The most common drug-related serious treatment-emergent adverse events were pneumonitis (three [4%]), ILD (two [3%]), and nausea (two [3%]).

As of the Nov 8, 2021, data cutoff, adjudicated ILD or pneumonitis related to trastuzumab deruxtecan occurred in eight patients (10%); two patients (3%) had grade 1 ILD or pneumonitis events, four (5%) had grade 2 events,

	April 9, 2021, data cutoff; patients (N=79)	Nov 8, 2021, data cutoff; patients (N=79)
Confirmed objective response	30 (38%; 27·3–49·6)	33 (42%; 30·8–53·4)
Confirmed best overall response		
Complete response	3 (4%)	4 (5%)
Partial response	27 (34%)	29 (37%)
Stable disease	34 (43%)	31 (39%)
Progressive disease	13 (16%)	13 (16%)
Not evaluable	2 (3%)	2 (3%)
Median progression-free survival, months	5·5 (4·2–7·2)*	5·6 (4·2–8·3)†
Patients with events	44 (56%)	51 (65%)
Progressive disease	37 (47%)	44 (56%)
Death	7 (9%)	7 (9%)
Median overall survival, months	12·1 (8·6–NE)‡	12·1 (9·4–15·4)§
Patients with events	26 (33%)	46 (58%)
Patients without events (censored)	53 (67%)	33 (42%)
Alive	46 (58%)	26 (33%)
Lost to follow-up	7 (9%)	7 (9%)
Confirmed disease control	64 (81%; 70·6–89·0)	64 (81%; 70·6–89·0)
Median time to response, months	1·4 (1·4–2·6)	1·4 (1·4–2·7)
Median duration of response, months	8·1 (4·1–NE)¶	8·1 (5·9–NE)‖

Data are n (%), estimate (%; 95% CI), or median (95% CI). NE=not estimable. *Assessed in the full analysis set based on 44 events. †Assessed in the full analysis set based on 51 events. ‡Assessed in the full analysis set based on 26 events. §Assessed in the full analysis set based on 46 events. ¶Assessed in responders (n=30); 20 patients were censored (because of initiating new anticancer therapy, event after missing two consecutive assessments, ongoing without event, or adequate tumour assessment no longer available). ‖Assessed in responders (n=33); 18 patients were censored (because of initiating new anticancer therapy, event after missing two consecutive assessments, ongoing without event, or adequate tumour assessment no longer available).

Table 2: **Summary of response assessment by independent central review, full analysis set**



B

	Patients, n	Complete responses and partial responses, n	Objective response rate (95% CI)
Region			
North America	34	10	29·4 (15·1–47·5)
Europe	45	23	51·1 (35·8–66·3)
Age			
<65 years	46	18	39·1 (25·1–54·6)
≥65 years	33	15	45·5 (28·1–63·6)
<75 years	75	29	38·7 (27·6–50·6)
≥75 years	4	4	100·0 (39·8–100·0)
Sex			
Female	22	12	54·5 (32·2–75·6)
Male	57	21	36·8 (24·4–50·7)
Eastern Cooperative Oncology Group performance status			
0	29	8	27·6 (12·7–47·2)
1	50	25	50·0 (35·5–64·5)
HER2 status			
IHC3+	68	32	47·1 (34·8–59·6)
IHC2+/ISH+	10	1	10·0 (0·3–44·5)
Primary tumour location			
Gastric	27	15	55·6 (35·3–74·5)
GEJ	52	18	34·6 (22·0–49·1)
Histological subtype			
Diffuse	1	1	100·0 (2·5–100·0)
Intestinal	19	7	36·8 (16·3–61·6)
Other	59	25	42·4 (29·6–55·9)
Number of metastatic sites			
<2	5	3	60·0 (14·7–94·7)
≥2	74	30	40·5 (29·3–52·6)
Previous adjuvant or neoadjuvant therapy			
Yes	9	3	33·3 (7·5–70·1)
No	70	30	42·9 (31·1–55·3)
Previous nivolumab or pembrolizumab treatment			
Yes	6	2	33·3 (4·3–77·7)
No	73	31	42·5 (31·0–54·6)
Previous treatment with immune-checkpoint inhibitor or other immuno-oncology therapy			
Yes	7	2	28·6 (3·7–71·0)
No	72	31	43·1 (31·4–55·3)
Liver metastases at baseline			
Yes	50	19	38·0 (24·7–52·8)
No	29	14	48·3 (29·4–67·5)
Renal impairment at baseline			
Normal	32	9	28·1 (13·7–46·7)
Mild	25	12	48·0 (27·8–68·7)
Moderate	8	5	62·5 (24·5–91·5)
Hepatic impairment at baseline			
Normal	64	26	40·6 (28·5–53·6)
Mild	14	7	50·0 (23·0–77·0)

one (1%) died due to ILD, and one (1%) died due to pneumonitis. The median time to onset of adjudicated drug-related ILD or pneumonitis was 80·5 days (IQR 56·5–148·0), with a median duration of 36·0 days (20·0–56·0).

On the basis of laboratory values of the left ventricular ejection fraction collected during the study, grade 2 events (defined as a resting left ventricular ejection fraction of at least 40% to less than 50%, or a 10% to less than 20% decrease from baseline) were observed in six (11%) of 57 patients. No clinical heart failure events were reported.

11 patients (14%) died, and two (3%) deaths were considered related to trastuzumab deruxtecan (both due to ILD or pneumonitis; appendix pp 7, 8). Deaths due to disease progression, neoplasm progression, and COVID-19 were reported in two patients each and were considered unrelated to trastuzumab deruxtecan. Deaths due to cerebrovascular accident, ILD, pneumonitis, intestinal obstruction, and lymphangiosis carcinomatosa were reported in one patient each. The two deaths due to ILD or pneumonitis were considered related to trastuzumab deruxtecan by the investigator and by the ILD adjudication committee (appendix pp 7, 8).

It should be noted that DESTINY-Gastric02 enrolled and treated patients throughout the COVID-19 pandemic. COVID-19 led to four patients interrupting their treatment and was attributed as the cause of death of two patients.

Discussion

The DESTINY-Gastric02 study showed clinically meaningful activity of trastuzumab deruxtecan, with durable responses in patients in the USA and Europe with HER2-positive gastric or gastro-oesophageal junction adenocarcinoma whose disease progressed on a trastuzumab-containing regimen. A high proportion of patients receiving trastuzumab deruxtecan in this study reached the primary endpoint of confirmed objective response rate by independent central review (33 [42%; 95% CI 30·8–53·4] of 79; Nov 8, 2021 data cutoff) compared with studies of other second-line treatments.[5,6,18,19] Furthermore, the activity outcomes of second-line therapy with trastuzumab deruxtecan in this study population are particularly promising when compared with historical data.

Figure 2: **Best percentage change of tumour size and objective resonse rate (full analysis set)**

(A) Best percentage change of tumour size from baseline by independent central review;* the red line at 20% indicates progressive disease and the black line at –30% indicates partial response. Analyses were done in the full analysis set. (B) Objective response rate across prespecified subgroups; two-sided 95% CIs are based on the exact (Clopper-Pearson) binomial distribution. The previous total gastrectomy (yes *vs* no) subgroup is not reported in part B as all patients were recorded as "no". *Three patients were missing baseline or postbaseline target lesion assessment and are excluded from Panel A. Nov 8, 2021, data cutoff.

Efficacy data from DESTINY-Gastric01, done in Japan and South Korea in the third-line or later setting, were similar, with a confirmed objective response rate of 42·0% (95% CI 33·0–51·4) with trastuzumab deruxtecan versus 12·5% (5·2–24·1) with physician's choice of irinotecan or paclitaxel (p=0·0001).[20] Median progression-free survival was 5·6 months with trastuzumab deruxtecan versus 3·5 months with physician's choice (hazard ratio [HR] 0·47 [95% CI 0·31–0·71]; p=0·0003), and median overall survival was 12·5 months with trastuzumab deruxtecan versus 8·9 months with physician's choice (HR 0·60 [95% CI 0·42–0·86]).[20]

There have been few studies in the second line in patients with HER2-positive gastric cancer, and the failure of previous trials for targeted therapies in this setting highlights the high unmet medical need.[5,6] Although the historical 27% objective response rate in the second-line setting we used in our sample size calculation was not specific to patients with HER2-positive gastric cancer, similar objective response rates have been reported in HER2-positive gastric cancer studies. In the randomised, open-label, phase 2/3 GATSBY study, 42 (21%) of 204 patients with previously treated, HER2-positive, advanced gastric cancer who received second-line trastuzumab emtansine 2·4 mg/kg weekly had an objective response, compared with 20 (20%) of 102 of those receiving taxane.[5] In the randomised, open-label, phase 3 TyTAN study, the objective response rate was 27% in patients with HER2-positive advanced gastric cancer who received second-line treatment with lapatinib plus paclitaxel versus 9% in patients who received paclitaxel alone (odds ratio [OR], 3·85 [95% CI 1·80–8·87]; p<0·001).[6] Finally, in the randomised, open-label, phase 2 T-ACT study, second-line treatment with paclitaxel plus trastuzumab achieved a similar response rate (33%) to that with paclitaxel alone (32%) in patients with HER2-positive advanced gastric or gastro-oesophageal junction cancer in Japan.[19]

A more useful comparator for the results of the present DESTINY-Gastric02 study might be the RAINBOW trial[18] of ramucirumab plus paclitaxel, which had positive results in the second-line setting. The patient population of these trials differed somewhat. The phase 2 DESTINY-Gastric02 study included 89 patients from the USA and Europe, the majority of whom were White (87%). In contrast, the phase 3 RAINBOW trial included 665 patients of a more diverse ethnic make-up, from 27 countries in North and South America, Europe, Asia, and Australia. Despite the differences, it is still useful to compare the two trials and noteworthy that second-line therapy with ramucirumab plus paclitaxel in the RAINBOW trial was associated with consistently lower response rates and survival outcomes compared with second-line trastuzumab deruxtecan in the DESTINY-Gastric02 trial. In the RAINBOW trial, objective response rate by investigator was 28% (95% CI 23–33) for ramucirumab plus paclitaxel versus 16% (13–20) for

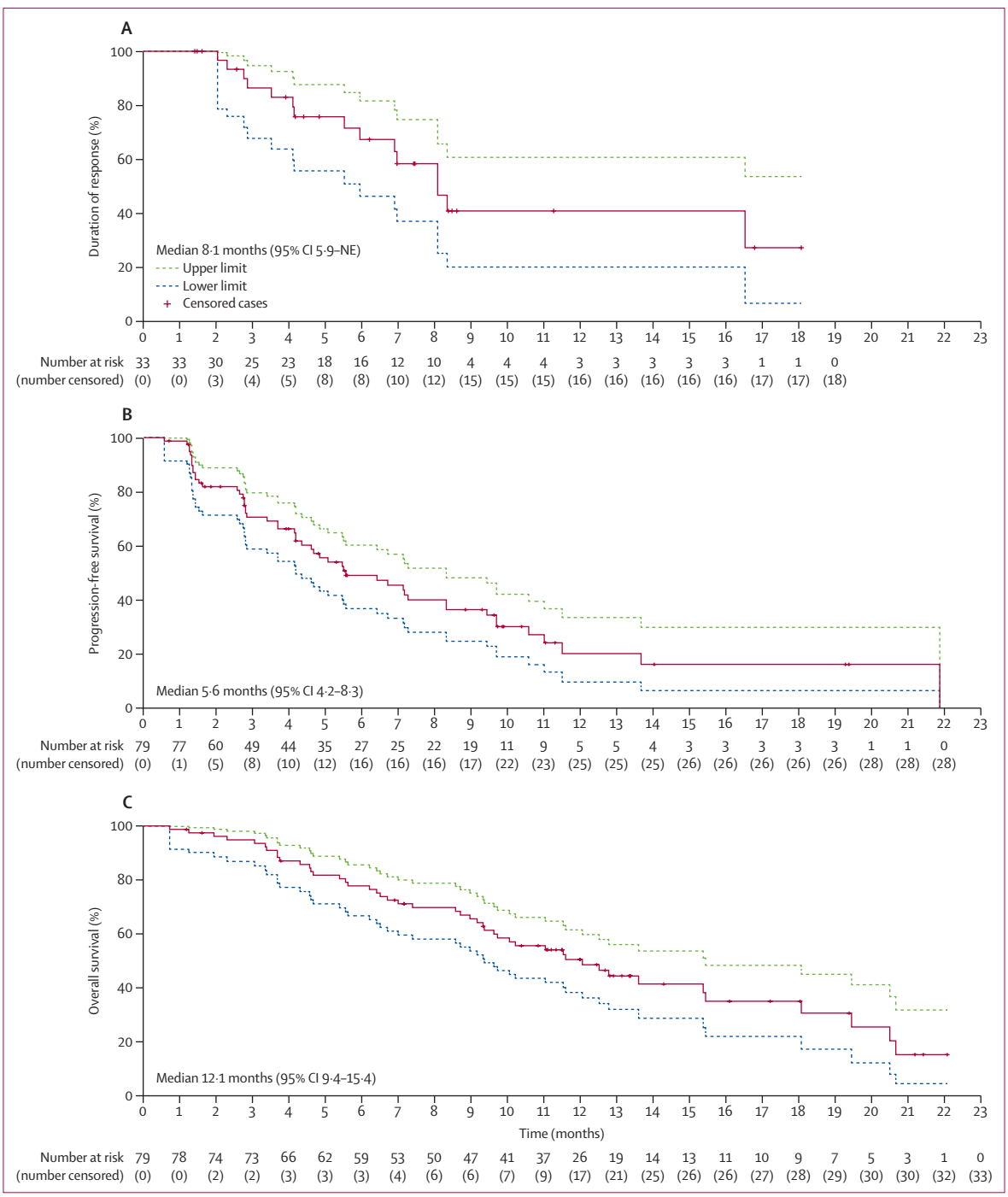


Figure 3: **Duration of response and progression-free survival by independent central review, and overall survival**
(A) Duration of response; analysis based on responders (n=33); 18 patients were censored for reasons including initiating new anticancer therapy, event after missing two consecutive assessments, ongoing without event, and adequate tumour assessment no longer available. (B) Progression-free survival; analysis done in the full analysis set based on 51 events (44 progressive disease, seven deaths). (C) Overall survival; analysis done in the full analysis set (N=79). Nov 8, 2021, data cutoff.

placebo plus paclitaxel (p=0·0001). Median progression-free survival with ramucirumab plus paclitaxel was 4·4 months (95% CI 4·2–5·3) versus 2·9 months (2·8–3·0) for placebo plus paclitaxel (HR 0·635 [95% CI 0·536–0·752]; p<0·0001), and median overall survival was 9·6 months (95% CI 8·5–10·8) versus 7·4 months (6·3–8·4; HR 0·807 [95% CI 0·678–0·962]; p=0·017).[18] In a subgroup analysis of patients in the RAINBOW trial who were from Western countries, including Europe, Israel, Australia, and the USA, objective response rate

by investigator (26·8%) and median overall survival (8·6 months) with ramucirumab plus paclitaxel were also lower than we found in DESTINY-Gastric02.[21]

Asian patients with gastric cancer participating in clinical trials typically have better survival outcomes than patients of White, Latinx, or Black ethnicity.[22,23] Potential explanations for the difference in gastric cancer epidemiology and treatment outcome between these populations include that distal gastric cancer is more common in Asian patients, whereas gastro-oesophageal junction cancer is more prevalent in patients from North America and Europe. Evidence suggests that Asian patients tend to have less advanced or aggressive disease at initial diagnosis than other populations, potentially due to gastric cancer screening programmes, and are more likely to receive systemic therapy,[22,23] and patients with gastric tumours tend to have better survival outcomes relative to gastro-oesophageal junction tumours.[24] Risk factors for gastro-oesophageal junction cancer include obesity, reflux disease, and low *Helicobacter pylori* infection rates, which are more common in Western countries.[25] However, Asian patients might be at a higher risk of ILD or pneumonitis (mild or moderate in severity) associated with trastuzumab deruxtecan compared with non-Asian patients, as suggested by a pooled analysis of nine clinical studies that compared rates of any-grade adjudicated drug-related ILD or pneumonitis in patients enrolled in Japan with those enrolled in other countries.[26] Notably, higher proportions of patients in DESTINY-Gastric02 had worse prognostic factors (such as two or more metastatic sites, presence of liver metastases, higher baseline tumour burden [as assessed by median sum of longest diameters], and an ECOG performance status of 1) than did those in the trastuzumab deruxtecan group of DESTINY-Gastric01.[1] The improved safety profile observed in patients from the USA and Europe in DESTINY-Gastric02 could potentially be attributed to the use of trastuzumab deruxtecan in an earlier-line treatment setting than in DESTINY-Gastric01.

In DESTINY-Gastric02, patients were required to have biopsy-confirmed HER2-positive disease. After treatment with trastuzumab, some patients have been observed to lose HER2 expression, potentially making their disease refractory to subsequent HER2-directed treatment.[6,27–29] HER2 stability during disease progression has not been widely reported, but HER2 loss has been reported to occur in about a third of patients in the few studies that re-evaluated HER2 status.[8,27] Studies to investigate whether HER2 expression is lost following treatment with trastuzumab deruxtecan are ongoing.

The safety profile of trastuzumab deruxtecan observed in DESTINY-Gastric02 was generally consistent with the established safety profile. The most common treatment-emergent adverse events were gastrointestinal or haematological in nature, and the incidence rates, including those of any grade or grade 3 or worse, and adverse

	Grade 1–2	Grade 3	Grade 4	Grade 5
Any treatment-emergent adverse event	35 (44%)	28 (35%)	5 (6%)	11 (14%)
Nausea	47 (59%)	6 (8%)	0	0
Vomiting	33 (42%)	2 (3%)	0	0
Fatigue	30 (38%)	3 (4%)	0	0
Diarrhoea	28 (35%)	1 (1%)	0	0
Decreased weight	25 (32%)	3 (4%)	0	0
Constipation	23 (29%)	0	0	0
Decreased appetite	22 (28%)	4 (5%)	0	0
Anaemia	19 (24%)	11 (14%)	0	0
Alopecia	19 (24%)	0	0	0
Decreased platelet count	12 (15%)	2 (3%)	0	0
Increased aspartate aminotransferase	12 (15%)	1 (1%)	0	0
Hypokalaemia	12 (15%)	0	1 (1%)	0
Abdominal pain	11 (14%)	2 (3%)	0	0
Asthenia	11 (14%)	1 (1%)	0	0
Pyrexia	9 (11%)	0	0	0
Blood alkaline phosphatase increased	8 (10%)	1 (1%)	0	0
Cough	8 (10%)	1 (1%)	0	0
Epistaxis	8 (10%)	0	0	0
Gastro-oesophageal reflux disease	8 (10%)	0	0	0
Hypoalbuminaemia	8 (10%)	0	0	0
Pneumonitis	7 (9%)	0	0	1 (1%)
Decreased neutrophil count	7 (9%)	2 (3%)	4 (5%)	0
Increased alanine aminotransferase	7 (9%)	1 (1%)	0	0
Dyspnoea	6 (8%)	1 (1%)	0	0
Hypotension	5 (6%)	1 (1%)	0	0
Neutropenia	4 (5%)	4 (5%)	0	0
Thrombocytopenia	4 (5%)	1 (1%)	0	0
Decreased white blood cell count	4 (5%)	5 (6%)	0	0
Device-related infection	3 (4%)	1 (1%)	0	0
Dysphagia	3 (4%)	3 (4%)	0	0
Ascites	3 (4%)	2 (3%)	0	0
Increased blood bilirubin	3 (4%)	2 (3%)	0	0
COVID-19	2 (3%)	1 (1%)	0	2 (3%)
Urinary tract infection	2 (3%)	2 (3%)	0	0
Hypophosphataemia	2 (3%)	2 (3%)	0	0
Hyperglycaemia	1 (1%)	1 (1%)	0	0
Hypertension	2 (3%)	1 (1%)	0	0
Interstitial lung disease	2 (3%)	0	1 (1%)	1 (1%)
Acute kidney injury	2 (3%)	2 (3%)	0	0
Cancer pain	1 (1%)	1 (1%)	0	0
Somnolence	1 (1%)	1 (1%)	0	0
Pleural effusion	1 (1%)	1 (1%)	0	0
Hepatotoxicity	1 (1%)	1 (1%)	0	0
Decreased lymphocyte count	1 (1%)	2 (3%)	0	0
Pneumonia	0	0	2 (3%)	0
Bacterial sepsis	0	0	1 (1%)	0
Catheter site infection	0	1 (1%)	0	0
COVID-19 pneumonia	0	1 (1%)	0	0
Staphylococcal infection	0	1 (1%)	0	0
Wound infection	0	1 (1%)	0	0
Malignant neoplasm progression	0	0	0	2 (3%)

(Table 3 continues on next page)

	Grade 1–2	Grade 3	Grade 4	Grade 5
(Continued from previous page)				
Lymphangiosis carcinomatosa	0	0	0	1 (1%)
Tumour haemorrhage	0	1 (1%)	0	0
Febrile neutropenia	0	1 (1%)	1 (1%)	0
Lymphopenia	0	1 (1%)	0	0
Electrolyte imbalance	0	1 (1%)	0	0
Basal ganglia infarction	0	1 (1%)	0	0
Cerebrovascular accident	0	0	0	1 (1%)
Generalised tonic-clonic seizure	0	1 (1%)	0	0
Embolism	0	2 (3%)	0	0
Pulmonary embolism	0	2 (3%)	0	0
Anal fissure	0	1 (1%)	0	0
Anal fistula	0	1 (1%)	0	0
Colitis	0	1 (1%)	0	0
Enteritis	0	1 (1%)	0	0
Haematemesis	0	1 (1%)	0	0
Obstruction gastric	0	1 (1%)	0	0
Intestinal obstruction	0	0	0	1 (1%)
Oesophageal obstruction	0	1 (1%)	0	0
Bile duct stenosis	0	1 (1%)	0	0
Gallbladder obstruction	0	1 (1%)	0	0
Hydronephrosis	0	1 (1%)	0	0
Urinary tract obstruction	0	1 (1%)	0	0
Disease progression	0	0	0	2 (3%)
Oedema	0	1 (1%)	0	0
Animal bite	0	1 (1%)	0	0
Exposure to communicable disease	0	1 (1%)	0	0
Product issues	0	1 (1%)	0	0
Device occlusion	0	1 (1%)	0	0

Data are n (%). Data are shown for adverse events that occurred in at least 10% (grade 1–2) of patients by worst toxicity grade and all grade 3–5 events, and listed in decreasing order of frequency. Nov 8, 2021, data cutoff.

Table 3: **Treatment-emergent adverse events, safety analysis set (n=79)**

events of special interest such as ILD or pneumonitis, were lower than those reported in DESTINY-Gastric01.[20]

Limitations of this study include the absence of a comparator or placebo group and a relatively small patient population. The small size of patient subgroups makes it difficult to draw more specific conclusions from the data. It is also worth noting that although a post-trastuzumab progression biopsy sample was needed for patients to be eligible for inclusion in this study, this might not always be feasible in clinical practice owing to the difficulty in obtaining adequate biopsy material or considering the clinical condition of the patient. Nevertheless, such a practice should be encouraged, on the basis of the potential loss of HER2 expression discussed in the introduction. Additionally, although this study is limited by its single-arm design, there have been relatively few studies in the second-line setting in patients with HER2-positive gastric cancer to date. DESTINY-Gastric02 provides data supporting clinically meaningful activity and a manageable safety profile and thus, compelling evidence that trastuzumab deruxtecan is a valuable second-line HER2-targeted treatment option for patients from the USA or Europe, similar to the findings of DESTINY-Gastric01 for Asian patients in the third-line and later settings. These findings have been validated by EU approval of trastuzumab deruxtecan for patients with gastric or gastro-oesophageal junction cancer who have received a previous trastuzumab-based regimen.[30] Planned analyses of this study will include evaluation of circulating biomarkers including cell-free DNA, an emerging potential diagnostic biomarker, and biomarker studies are ongoing to evaluate the use of liquid biopsies in gastric cancer and other solid tumours. Results from DESTINY-Gastric02 also support the ongoing randomised, phase 3 DESTINY-Gastric04 (NCT04704934) trial, which is investigating the efficacy and safety of trastuzumab deruxtecan compared with ramucirumab and paclitaxel in patients with gastric or gastro-oesophageal junction cancer whose disease has progressed on or after a trastuzumab-containing regimen.

Contributors

EVC, FB, YK, AQ, JS, and GK contributed to study conception and design. EVC, MdB, ES, IC, HP, SS, SL, ZAW, JA, JC, and GK were principal investigators and contributed to study initiation. YJ also contributed to study initiation. FB, YK, AQ, and JS contributed to data interpretation. All authors participated in data acquisition and quality control, had full access to all data from the study, and approved the manuscript. YK and GK accessed and verified the underlying study data.

Declaration of interests

EVC has received research grants paid to his institution from Amgen, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Celgene, Ipsen, Lilly, Merck Sharp & Dohme, Merck, Novartis, Roche, and Servier and has participated in advisory boards for AbbVie, Array, Astellas, AstraZeneca, Bayer, Beigene, Biocartis, Boehringer Ingelheim, Bristol Myers Squibb, Celgene, Daiichi Sankyo, Halozyme, GlaxoSmithKline, Helsinn, Incyte, Ipsen, Janssen Research, Lilly, Merck Sharp & Dohme, Merck, Mirati, Novartis, Pierre Fabre, Roche, Seagen, Servier, Sirtex, Terumo, Taiho, TRIGR, and Zymeworks. ES has received consulting fees from Amal Therapeutics, AstraZeneca, Bristol Myers Squibb, Beigene, Daiichi Sankyo, Merck, Novartis, Pfizer, Roche, Servier, Turning Point Therapeutics, and Zymeworks; speaker payments or honoraria from Amgen, Bristol Myers Squibb, Novartis, and Servier; support for meeting attendance or travel from Amgen, Bristol Myers Squibb, and Servier; and participated on a data safety monitoring board or advisory board for Amgen, AstraZeneca, and Beigene; and held a leadership role (co-chair) of the EORTC GI Trials Group Gastric Cancer Task Force. IC has received research funding from Eli Lilly and Janssen-Cilag, has participated in advisory boards for Eli Lilly, Bristol Myers Squibb, Merck Sharp & Dohme, Roche, Merck Serono, AstraZeneca, OncXerna, Pierre Fabre, Boehringer Ingelheim, Incyte, Astellas, GlaxoSmithKline, Sotio, Eisai, Daiichi Sankyo, Taiho, Servier, Seagen, and Turning Point Therapeutics; speaker payments or honoraria from Eli Lilly, Eisai, and Servier; and has received support for meeting attendance or travel from Bristol Myers Squibb. HP has received grants or contracts from Adlai Nortye USA, Alpine Immune Sciences, Ambrx, Amgen, Aprea Therapeutics, Array BioPharma, Bayer, BeiGene, BJ Bioscience, Bristol Myers Squibb, Daiichi Sankyo, Eli Lilly, Elicio Therapeutics, EMD Serono, Exelixis, Genentech, Gilead Sciences, GlaxoSmithKline, Gossamer Bio, Hoffman-LaRoche, Hutchison MediPharma, ImmuneOncia Therapeutics, Incyte, Jounce Therapeutics, Mabspace Biosciences, MacroGenics, Medimmune, Medivation, Merck, Millennium, Mirati Therapeutics, Novartis Pharmaceuticals, Oncologie, Pfizer, PsiOxus Therapeutics, Puma Biotechnology, Regeneron Pharmaceuticals, RePare Therapeutics, Seagen, Synermore Biologics, Taiho Pharmaceutical, TopAlliance Biosciences, Turning Point Therapeutics, Vedanta Biosciences, Xencor,

and participated on a data safety monitoring board or advisory board for Jacobio. SS has participated in advisory boards for Agenus, AstraZeneca, Bayer, Bristol Myers Squibb, CheckmAb, Daiichi Sankyo, Guardant Health, Menarini, Merck, Novartis, Pierre Fabre, Roche-Genentech, and Seagen. SL has received research funding paid to her institution from Daiichi Sankyo, Amgen, Merck Serono, Bayer, Roche, Lilly, AstraZeneca, and Bristol Myers Squibb; consulting fees from Amgen, Merck Serono, Lilly, AstraZeneca, Incyte, Daiichi Sankyo, Bristol Myers Squibb, Servier, and Merck Sharp & Dohme; and speaker payments or honoraria from Roche, Lilly, Bristol Myers Squibb, Servier, Merck Serono, Pierre Fabre, GlaxoSmithKline, and Amgen. ZAW has received consulting fees from Merck, Ibsen, Lilly, Five Prime, QED, Molecular Templates, Daiichi Sankyo, AstraZeneca, Bayer, and Novartis; support for meeting attendance or travel from Lilly, Merck, Novartis, and Daiichi Sankyo; and participated on a data safety monitoring board or advisory board for Array. JC has received research funding paid to his institution from Daiichi Sankyo, Merck, and Brooklyn Immunotherapeutics; consulting fees from Lilly, Merck, AstraZeneca, Foundation Medicine, Daiichi Sankyo, Macrogenics, Amgen, Ono Pharmaceutical, Bristol Myers Squibb, Astellas, Turning Point Therapeutics, Silverback Therapeutics, Novartis, Coherus Biosciences, Geneos, and Roche; speaker payments or honoraria from Merck and Bristol Myers Squibb; and participated on a data safety monitoring board for Yiviva. YJ has received grants or contracts from Bayer, Bristol Myers Squibb, Cycle for Survival, Department of Defense, Eli Lilly, Fred's Team, Genentech Roche, Merck, NCI, and RGENIX; speaker payments or honoraria from Amerisource Bergen, Arcus Biosciences, AstraZeneca, Basilea Pharmaceutica, Bayer, Bristol Myers Squibb, Daiichi Sankyo, Eli Lilly, Geneos Therapeutics, GlaxoSmithKline, Imedex, Imugene, Lynx Health, Merck, Merck Serono, Michael J Hennessy Associates, Paradigm Medical Communications, PeerView Institute, Pfizer, Research to Practice, RGENIX, Seagen, Silverback Therapeutics, and Zymeworks; and stock options from RGENIX. FB was an employee at the time of this study and held stocks or stock options in Daiichi Sankyo. YK, AQ, and JS are employees of Daiichi Sankyo. GK has received research funding paid to his institution from Arog, AstraZeneca, Bristol Myers Squibb, Daiichi Sankyo, Merck, Oncolys, Pieris, and Zymeworks; consulting fees from Apexigen, AstraZeneca, Bristol Myers Squibb, Merck, Pieris, and Zymeworks; support for meeting attendance or travel from Bristol Myers Squibb, Merck, and Pieris; and participated on a data safety monitoring board or advisory board for Zymeworks. MdB and JA declare no competing interests.

Data sharing

Anonymised individual participant data on completed studies and applicable supporting study documents may be available upon request at https://vivli.org/. In cases where clinical study data and supporting documents are provided pursuant to our company policies and procedures, Daiichi Sankyo Companies will continue to protect the privacy of the company and our clinical study participants. Details on data sharing criteria and the procedure for requesting access can be found on our website. More information on data sharing is in the appendix (p 2).

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